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Exopolysaccharide production by the cyanobacterium *Anabaena* sp. ATCC 33047 in batch and continuous culture

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Abstract

The halotolerant, filamentous, heterocystous cyanobacterium *Anabaena* sp. ATCC 33047 released, during the stationary growth phase in batch culture and, at low dilution rate, in continuous culture, large amounts of an exopolysaccharide (EPS) to the culture medium. Different environmental, nutritional and physical parameters affected production and accumulation of the EPS. The presence of either a combined nitrogen source or NaCl at high concentration led to decreased EPS production, without affecting cell growth. In contrast, generation of the EPS was markedly enhanced in response to an increase in either air flow rate, temperature or irradiance. In continuous culture, accumulation of EPS in the medium increased in response to a decrease in the dilution rate, with maximal EPS productivity being reached at a dilution rate of 0.03 h^{-1}. © 1998 Elsevier Science B.V. All rights reserved.

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1. Introduction

Microbial exopolysaccharides (EPS) are attracting increasing attention, since they represent interesting alternatives to polysaccharides of plant or macroalgal origin, traditionally used in the food, textile, painting, cosmetic, paper and pharmaceutical industries as emulsifiers, stabilizers or thickening agents. Some of these microbial EPS exhibit unique physical properties, which can be advantageous for new applications. In addition, as producers, microorganisms are in principle, better suited than macroalgae or higher plants, exhibiting higher growth rates and being more amenable to manipulation of conditions for enhancing

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growth and/or EPS production (Anderson and Eakin, 1986; Bertocchi et al., 1990; Sutherland, 1990).

Cyanobacteria (blue-green algae) are a well-known biological source of chemicals of commercial value, such as, among others, phycobiliproteins, carotenoids and fatty acids (Rodríguez et al., 1989; Rodríguez and Guerrero, 1992; Cohen et al., 1993; Moreno et al., 1995). They are photosynthetic prokaryotes present in virtually every ecosystem (Carr and Whiton, 1982). Many strains are filamentous and diazotrophic, i.e. they can use atmospheric dinitrogen, thus not requiring combined nitrogen as a nutrient. Besides lowering the cost of the culture medium, this ability helps to alleviate the problem of contamination by other microorganisms. Moreover, the filamentous nature of these microorganisms is an advantage for harvesting the cells (Fontes et al., 1987).

Several cyanobacteria produce exocellular mucilaginous material, mainly of polysaccharidic nature (Bertocchi et al., 1990). It is generally accepted that the synthesis of exocellular polysaccharides represents a metabolic strategy for survival and growth of cyanobacteria, especially under unfavourable environmental conditions (Flaibani et al., 1989; Vincenzini et al., 1990). Recent studies on the physical properties of cyanobacterial EPS show interesting flocculating and gelling properties, which may be exploited in either conventional or novel industrial applications. Among alternative uses of cyanobacterial EPS are the conditioning of soils for the improvement of their water-holding capacity, waste-water management, detoxification of metal-contaminated media, etc. (Barclay and Lewin, 1985; Flaibani et al., 1989; Bertocchi et al., 1990; Bender et al., 1994).

Although reports have recently appeared concerning the release by several strains of cyanobacteria of polymeric material, generally of polysaccharidic nature, most of them deal mainly with chemical composition studies, and only a few address the kinetics and ecophysiology of polysaccharide production (Flaibani et al., 1989; Bertocchi et al., 1990; Bender et al., 1994).

Anabaena sp. ATCC 33047 is a halotolerant, allophycocyanin-rich, filamentous, heterocystous, nitrogen-fixing cyanobacterium. It exhibits high salt tolerance, its growth only slightly affected by the presence of NaCl up to 0.5 M in the culture medium (Stacey et al., 1977; Moreno et al., 1995).

Production of exocellular polysaccharides by microalgae and cyanobacteria is known to respond to changes in several external factors, such as temperature, nitrogen concentration or irradiance (Kroen and Rayburn, 1984; Thepenier and Gudin, 1985; De Philippis et al., 1991; Lupi et al., 1991). The influence of some physical, environmental and nutritional factors on exopolysaccharide production by *Anabaena* sp. ATCC 33047 in batch and continuous culture has been tested with the aim of optimizing the production of EPS by this cyanobacterium.

2. Materials and methods

2.1. Culture conditions

Anabaena sp. ATCC 33047 was grown at 40°C (except in the experiments aimed to study the effect of temperature) on a medium containing 50 mM $NaHCO_3$, 8 mM KCl, 1 mM K_2HPO_4, 0.5 mM $MgSO_4$, 0.35 mM $CaCl_2$ and 85 mM NaCl (except in the experiments aimed to study the effect of NaCl concentration), as well as a supply of essential micronutrients and Fe-EDTA (Arnon et al., 1974).

Batch culture was carried out in 5 cm deep 1 l capacity Roux flasks, bubbled with air (250 l l^{-1} h^{-1}, except in the experiments aimed to study the effect of air flow rate) supplemented with 6% (v/v) CO_2 as the source of carbon and nitrogen. Flasks were laterally and continuously illuminated with mercury halide lamps at a surface irradiance of 460 $\mu E\ m^{-2}\ s^{-1}$ (except in the experiments aimed to study the effect of irradiance).

Continuous culture was performed in a cylindrical photobioreactor (5 l capacity, 16 cm internal diameter and 25 cm height), provided with temperature, pH, O_2 and CO_2 probes, and computerized control and data acquisition (Braun-Biotech, Biostat MD). Turbulence was provided with a rotating paddle (300 rpm) and air bubbling at a flow rate of 18 l l^{-1} h^{-1}. The cell suspension was kept at pH 8 through intermittent addition of pure CO_2, governed by a pH-stat system. Illumination was supplied by three halogen Sylvania 300 W lamps giving an average surface irradiance of 2500 μE m^{-2} s^{-1}, and regulated by rheostats.

2.2. Polysaccharide assay

Exopolysaccharide (EPS) released to the medium by *Anabaena* sp. ATCC 33047 was determined by the method described by Jarman et al. (1978). Aliquots (15–25 ml) of cell suspension were heated at 50°C for 10 min in the presence of 0.3 M NaCl and 0.03 M Na-EDTA, pH 10, and centrifuged at $27\,000 \times \underline{g}$ for 45 min at room temperature. EPS in the supernatant was precipitated by addition of 3 vol. of 2 propanol, collected by filtration through GF/C Whatman glass microfiber filters, washed with 2 propanol:water (3:1), and dried at 80°C to constant weight.

2.3. Nitrogenase assay

Cellular level of nitrogenase activity was estimated as acetylene reducing activity (Stewart et al., 1967).

2.4. Analytical procedures

Biomass dry weight determination was carried out as described by Fontes et al. (1987). Phycobiliproteins were estimated spectrophotometrically, employing the equations given by Siegelman and Kycia (1978).

3. Results

3.1. EPS production in batch culture

Values of specific growth rate and EPS production by *Anabaena* sp. ATCC 33047 grown with different nitrogen sources (N_2, KNO_3 or NH_4Cl) in batch regime are shown in Table 1. Cell growth under nitrogen-fixing conditions was similar to that recorded in the presence of combined nitrogen. Nevertheless, high EPS production was observed only under diazotrophic conditions.

Fig. 1 shows the kinetics of growth, nitrogenase level and EPS production by *Anabaena* sp. ATCC 33047 in batch culture. EPS concentration in the culture increased slowly during the exponential phase of growth. Once the stationary phase was reached, however, EPS accumulation increased significantly, reaching a maximum of ~ 17 g l^{-1} after 5–6 days from the beginning of growth. The massive release of this polysaccharide into the medium increased culture viscosity with age. Cellular nitrogenase activity was maximal during the exponential growth phase, when EPS production was low, but decreased rapidly shortly thereafter, to reach a minimal level, coincident with the enhancement in EPS production. Thus, nitrogen starvation, induced by low nitrogen fixation capacity of the cells, appears to be a requirement for high EPS production rate. Actually, cells in the stationary growth phase, actively producing EPS, exhibited much lower phycobiliprotein levels (12% of dry weight) than cells growing exponentially (28% of dry weight), again indicating that the former cells were suffering from nitrogen stress.

The presence of added NaCl had a marked effect on EPS production by *Anabaena* sp. ATCC 33047 cells growing under diazotrophic conditions (Table 2). Maximal EPS production values were

Table 1
Effect of the nitrogen source on specific growth rate and EPS production by *Anabaena* sp. ATCC 33047 in batch culture

Nitrogen source	Specific growth rate (d−1)	Exopolysaccharide production (g l−1)
N_2 (air)	1.87 ± 0.2	17.2 ± 1.4
KNO_3 (20 mM)	2.26 ± 0.2	2.0 ± 0.1
NH_4Cl (5 mM)	1.75 ± 0.1	2.7 ± 0.2

Cells were grown under standard culture conditions, either in the absence or in the presence of a combined nitrogen source (KNO_3 or NH_4Cl). The stated concentrations of KNO_3 and NH_4Cl in the medium were adjusted accordingly, daily. Specific growth rate values were determined in the logarithmic phase. EPS production values correspond to those measured in the stationary phase of growth (fifth day of culture growth). Data are mean values ± S.D. of three independent measurements.

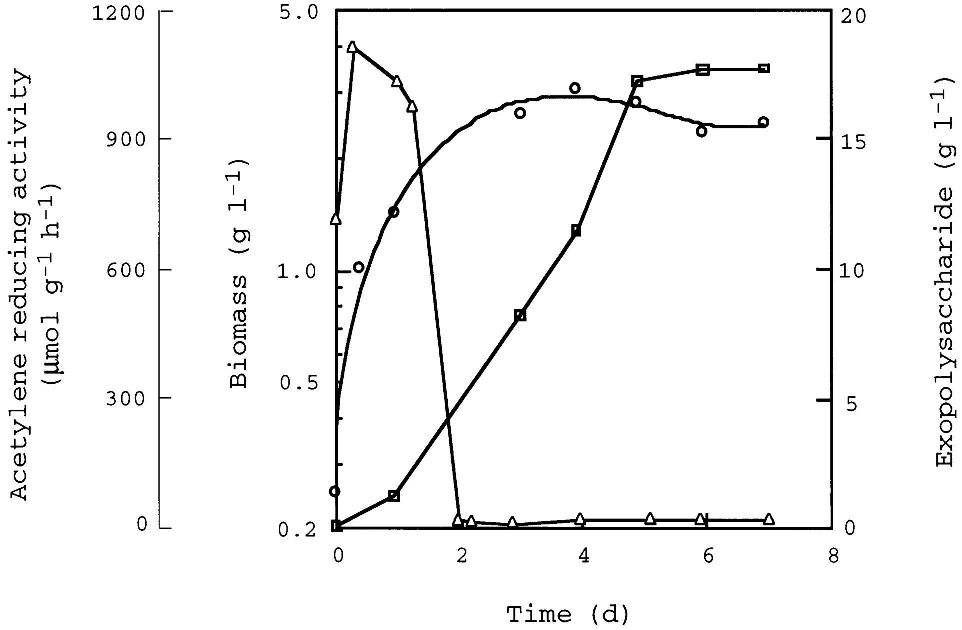

Fig. 1. Time course of growth, EPS production and nitrogenase activity (expressed as acetylene reducing activity) of *Anabaena* sp. ATCC 33047 cells in batch culture. Cells were grown under standard culture conditions in the absence of combined nitrogen. Data are mean values of three independent measurements, S.D. < 10%. Symbols: ○, biomass; □, exopolysaccharide; △, acetylene reducing activity.

recorded in the absence of added NaCl and decreased markedly as NaCl in the medium was increased. During the stationary growth phase, cells grown in the presence of 85 mM NaCl exhibited both phycobiliproteins and nitrogenase levels (11% of dry weight and 2 μmol C_2H_4 g^{-1} h^{-1}, respectively) lower than those of cells grown at 340 mM NaCl (27% and 250 μmol C_2H_4 g^{-1} h^{-1}, respectively). These results further indicate, that EPS production by *Anabaena* sp. ATCC 33047 occurs significantly, only under conditions in which the nitrogenase activity level and phycobiliprotein content of the cells are low.

Table 2
Effect of NaCl concentration on EPS production by *Anabaena* sp. ATCC 33047 in batch culture

NaCl (mM)	Exopolysaccharide production (g 1−1)
0	13.5 ± 1.1
85	9.2 ± 0.6
170	7.9 ± 0.6
340	3.9 ± 0.2
425	3.5 ± 0.2
510	3.9 ± 0.3

The culture medium was supplemented with NaCl at the indicated concentrations. Data are mean values ± S.D. of three independent measurements in the stationary phase of growth (sixth day of culture time). Other experimental conditions were as in Fig. 1.

Release of EPS was also affected by temperature, being low (3 g l^{-1}) at 30–35°C, and increasing markedly (~ 4–5-fold) when the temperature was raised to 40–45°C. At 40–45°C, exponential growth of this cyanobacterium was faster and the time required to reach the onset of stationary phase was shorter than at 30–35°C (data not shown).

Fig. 2 shows the effect of irradiance on EPS production by *Anabaena* sp. ATCC 33047. At irradiance values between 115 and 345 μE m^{-2} s^{-1}, EPS production was ≈ 3 g l^{-1}. Increasing irradiance to 460 μE m^{-2} s^{-1} induced a 4-fold increase in EPS production, without further enhancement at higher irradiance values. Therefore, EPS production was light-saturated at 460 μE m^{-2} s^{-1}.

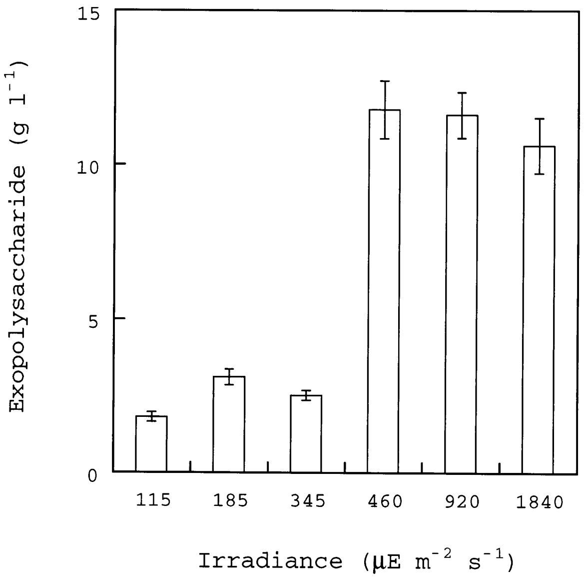

Fig. 2. Effect of irradiance on EPS production by *Anabaena* sp. ATCC 33047 in batch culture. Cultures were illuminated at the indicated irradiances. EPS was determined in the stationary phase of growth (fifth day of culture time). Vertical bars represent S.D., $n = 3$. Other experimental conditions were as in Fig. 1.

Fig. 3 shows the effect of air flow rate (a factor which, under the conditions used, determines the turbulence of the culture) on growth and EPS production by *Anabaena* sp. ATCC 33047. Increasing air flow rate from 50 to 250 $1\,l^{-1}\,h^{-1}$ did not affect cell growth, with both growth rate during exponential phase and maximal cell density values being similar and virtually unaffected by turbulence. Nevertheless, EPS accumulation increased markedly (from 4–5 to 14–15 g l^{-1}) as air flow rate was raised from 150 to 250 $1\,l^{-1}\,h^{-1}$. Therefore, EPS production during the stationary phase is strongly affected by aeration.

3.2. EPS production in continuous culture

Production of EPS has also been tested under continuous culture conditions, in a photobioreactor. Fig. 4 shows the effect of dilution rate on standing biomass, EPS concentration and EPS productivity in *Anabaena* sp. ATCC 33047 continuous cultures. The highest values of standing biomass and EPS concentration were obtained at the lowest dilution rate tested (0.013 and 0.03 h^{-1}), which corresponded to low specific growth rate. In contrast, when the dilution rate was increased (high specific growth rate), both standing biomass and EPS decreased. These results agree with those obtained in batch culture, with the highest EPS release being recorded at high cell density values. It is worth recalling that, in absolute terms, generation of polysaccharide was significantly higher than that of cell biomass, with

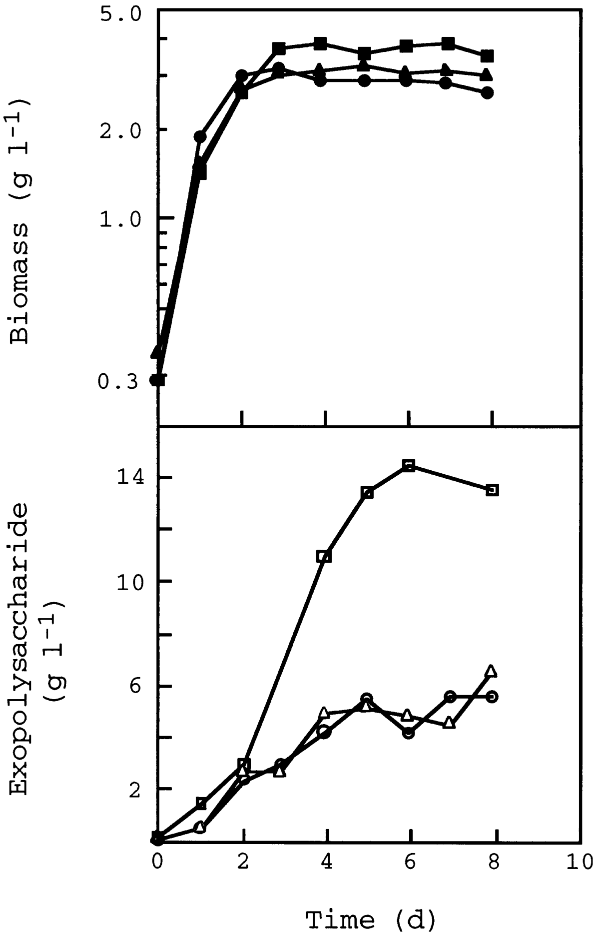

Fig. 3. Effect of air flow rate on growth and EPS production by *Anabaena* sp. ATCC 33047 in batch culture. Cell suspensions were sparged with 6% (v/v) CO_2-supplemented air at the indicated flow rates. Symbols: ○ and ●, 50; △ and ▲, 150; □ and ■, 250 $1\,l^{-1}\,h^{-1}$, respectively. Data are mean values of three independent measurements, S.D. < 10%. Other experimental conditions were as in Fig. 1.

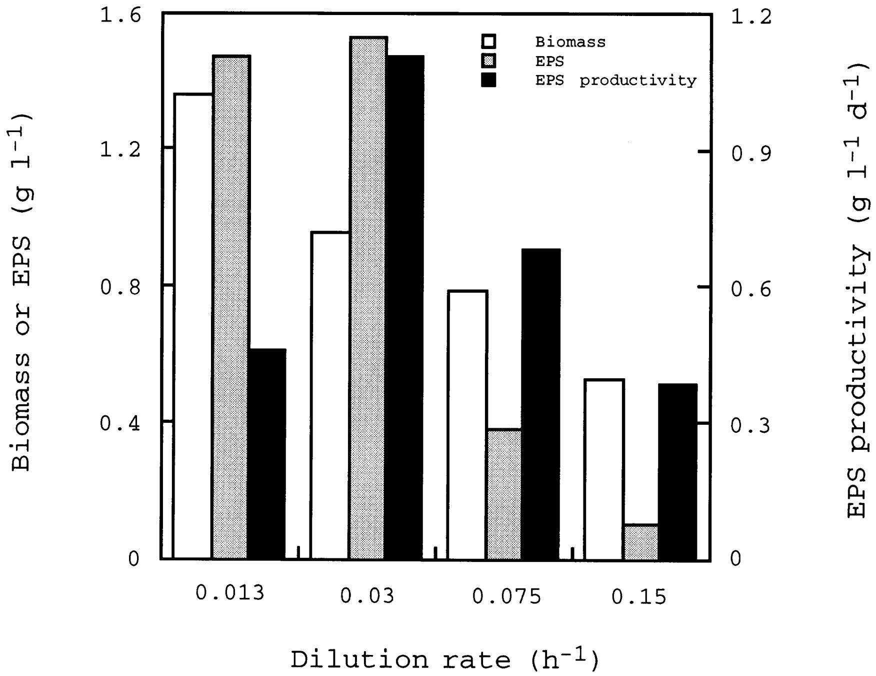

Fig. 4. Effect of dilution rate on growth, EPS concentration and EPS productivity by *Anabaena* sp. ATCC 33047 in continuous culture. Cells were grown at the indicated dilution rate, in the absence of a combined nitrogen source. Data are mean values of three independent measurements, S.D. < 10%.

the EPS/biomass ratio reaching values > 1.5 (Fig. 4, dilution rate, 0.03 h^{-1}). EPS productivity reached a maximal value of ~1.1 g l^{-1} d^{-1} at a dilution rate of 0.03 h^{-1}. At higher or lower dilution rate values, the EPS productivity decreased considerably.

4. Discussion

Optimal conditions for EPS production by *Anabaena* sp. ATCC 33047 do not coincide with those for cell growth. In fact, EPS release takes place mainly during the stationary phase, in which growth is limited by different factors. This behaviour agrees with the situation found for the microalgae *Chlamydomonas mexicana* and *Porphyridium* sp. (Kroen and Rayburn, 1984; Arad et al., 1988). In contrast, in the cyanobacterium *Cyanospira capsulata* and the green colonial microalga *Botryococcus braunii*, EPS was continuously released into the medium throughout the culture period (Vincenzini et al., 1990; Lupi et al., 1994).

Changes in some nutritional, environmental or physical factors influenced the synthesis and release of EPS by *Anabaena* sp. ATCC 33047, sometimes affecting cell growth in a different manner. Thus, although growth of this cyanobacterium was similar under diazotrophic conditions and in the presence of a combined nitrogen source, high EPS production only occurred in the first case. Similar results have been reported for the non-nitrogen fixers *Porphyridium* sp., *C. mexicana* and *B. braunii*, in which EPS production was inversely correlated with the amount of combined nitrogen present in the medium (Kroen and Rayburn, 1984; Arad et al., 1988; Lupi et al., 1994). EPS production by *Anabaena* sp. ATCC 33047 appears to be particularly enhanced under conditions of nitrogen stress, with reduced levels of nitrogenase activity and phycobiliprotein content (Fig. 1). The signal for increased polysaccharide synthesis seems to appear coincidentally with the end of the exponential growth phase, being otherwise linked to nitrogen starvation, and might be an integral part of the metabolic changes that occur within the cells at that time.

In relation to a possible ecological function of EPS release by *Anabaena* sp. ATCC 33047, there are some reports dealing with the ecological role of polysaccharide release by microalgae and cyanobacteria (Barclay and Lewin, 1985; Flaibani et al., 1989; Bender et al., 1994). For the cyanobacterium *Microcystis flos-aquae* C3-40, an interaction between its capsular polysaccharide and metal ions has been proposed to play a role in the environmental cycling of certain multivalent metals, especially manganese, during *Microcystis* blooms in many aquatic environments (Parker et al., 1996). Whether such specific interactions between metal ions and the EPS release of *Anabaena* sp. ATCC 33047 exist, remains to be established.

Enhancement of EPS production by *Anabaena* sp. ATCC 33047 at its optimal growth temperature (40–45°C) and at irradiance values $> 460\ \mu E\ m^{-2}\ s^{-1}$ can be explained in terms of a faster onset of the stationary phase under these conditions, with the subsequent earlier start of the EPS-releasing phase.

Increased turbulence did not affect cell growth, but resulted, however, in enhanced EPS production by *Anabaena* sp. ATCC 33047 ($\sim$ 3-fold, Fig. 3). This would suggest that turbulence facilitates release of the polysaccharide from the cell surface, thus stimulating synthesis of new exopolysaccharide. This positive effect of the turbulence on EPS release was also recorded in continuous culture at low dilution rate (data not shown).

The behaviour of *Anabaena* sp. ATCC 33047 with regard to EPS production has also been characterized under continuous culture conditions in a photobioreactor. The highest accumulation of EPS in the medium occurred at low dilution rate, with maximal productivity at a value of $0.03\ h^{-1}$ (Fig. 4). Under these conditions, cells are light-limited, with low nitrogenase and photosynthetic activity. These results agree with those obtained in batch culture, in which higher EPS release corresponded to higher cell density.

The recorded values for EPS production by *Anabaena* sp. ATCC 33047 are remarkably higher than those reported for other microalgae, including cyanobacteria (Anderson and Eakin, 1986; Bertocchi et al., 1990). Rheological studies of the EPS released by this cyanobacterium showed a pseudoplastic behaviour and a high apparent viscosity, similar to that of some currently used commercial gums (Madiedo et al., 1994). The high productivity values obtained, together with the interesting rheological properties of the *Anabaena* sp. ATCC 33047 EPS may allow exploitation of this organism as a polysaccharide producer. Furthermore, the lack of requirement of combined nitrogen and the high optimal growth temperature, together with the fast growth rate of *Anabaena* sp. ATCC 33047 give advantages for operations under outdoor conditions. In addition, considerable amounts of commercially valuable chemicals, such as phycobiliproteins (Moreno et al., 1995), are present in the cell residues remaining after EPS removal. Thus, *Anabaena* sp. ATCC 33047 cultures appear highly suitable for carrying out a multi-product strategy, with a rational utilization of some of their most interesting, intracellular and extracellular compounds.

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